

Rahul Rawat

IT DATENVISUALISIERUNG WERKSTUDENT

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PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on dashboarding, data visualisation, and production data work. I have shipped an interactive Tableau dashboard with dynamic Set Actions and parameter driven analytics, a fully automated BigQuery medallion pipeline feeding a five page Looker Studio dashboard, and a real time flight tracking pipeline processing over 128 thousand records on Google Cloud. Comfortable in Python, SQL, dashboarding tools, and structured data preparation, I am the right fit for building and maintaining production and machine data dashboards on your central data platform.

EXPERIENCE

eRay GmbH

Oct 2025 to Mar 2026

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

EDUCATION

M.Sc. Data Science and Analytics

Apr 2025 to Present

SRH University of Applied Sciences Heidelberg

Bachelor of Technology in Computer Science

2019 to 2023

GL Bajaj Institute of Technology and Management

PERSONAL PROJECTS

Fast Food Nutritional Analyzer and Meal Simulator

Tableau with Set Actions

parameters and calculated fields

data storytelling and UI or UX

- Built a dynamic shopping cart inside Tableau, measured by the fact that end users can select scatter plot points and instantly total calories, fat, and protein for a simulated meal, using Tableau Set Actions as the interaction layer.
- Implemented parameter driven analytics with a dynamic Y axis tied to a user controlled goal parameter for muscle gain versus weight loss, measured by consistent axis switching without dashboard reloads, using a CASE statement over the parameter.
- Authored complex order of operation IF and THEN calculated fields for logical grouping and custom flags such as an Is It A Trap flag for deceptive high fat and high calorie items, measured by manual spot checks against the source nutrition data.
- Designed a two tier layout combining an executive macro view and a granular food finder using a colour blind safe dark mode palette, measured against a target of reduced time to insight for a non technical stakeholder.

Movie Analytics and ML Pipeline on GCP

GCP BigQuery

Cloud Run

GCS

Cloud Scheduler

BigQuery ML

Python

SQL

Looker Studio

- Built an end to end batch pipeline pulling movie data from a public API into a GCS data lake and processing it through Bronze to Silver to Gold medallion architecture in BigQuery, measured by a fully automated Cloud Scheduler trigger with no manual intervention required, using Cloud Run for compute.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication via window functions, and genre normalisation into a relational model, measured by clean referential integrity across all downstream Gold tables.
- Trained a BigQuery ML classifier that predicts whether a film will be a hit before release, measured by leakage free evaluation, by deliberately splitting features into two tables so only pre release signals feed the model.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions on genre ROI, foreign language growth, and release season timing, measured against the ML early warning view, and secured the system with a least privilege service account and Secret Manager.

Real Time Flight Tracking Data Pipeline

Python

PySpark

BigQuery

dbt

Apache Airflow

GCP Dataproc and GCS

Tableau with TabPy

- Collected live flight positions from the OpenSky Network API every 30 seconds and enriched them against airport, aircraft, and weather data across four sources, measured by a clean joined table covering more than 128 thousand records, using Python collectors and PySpark cleaning on Google Cloud.
- Shaped raw collector output into analysis ready tables with dbt and computed each aircraft's nearest airport along the way, measured by consistent nearest airport labels across the historical dataset, using PySpark for heavy lift and dbt for the modelling layer.
- Orchestrated the whole system with Apache Airflow so batch and real time layers refresh automatically every 15 minutes, measured by unattended DAG runs on GCS backed storage and Dataproc compute.
- Surfaced findings in a Tableau workbook backed by Python statistics through TabPy, measured by the finding that air traffic drops 4.4 times in heavy rain and clusters around hubs like Frankfurt and Munich, using dbt aggregates as the visual layer feed.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 patients, using 10 fold cross validation and per model confusion matrices to make the model comparison auditable.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity before any model fit.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns in the minority class.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study, publishable in substance.

CERTIFICATIONS

- SAS Certified Specialist: Visual Business Analytics Using SAS Viya, issued May 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English: fluent, professional working proficiency. **German:** B1, in progress.